

TIC TAC TOE GAME

Project Statement

The **Tic Tac Toe Game** is a simple Python-based two-player console game developed as a programming project. The main purpose of this project is to demonstrate the use of basic Python programming concepts such as variables, conditional statements, loops, functions, lists, strings, user input, and file handling.

The game allows two players to enter their names and play against each other using the symbols **X** and **O**. Players select positions from **1 to 9** on the game board, and the program checks the selected positions to determine whether a player has completed a winning combination horizontally, vertically, or diagonally.

The project also includes a menu-based system through which users can start a new game, view the current score, view previous game records, display player details, reset the score and stored records, or exit the program.

File handling is used to store game results in a text file. The program creates a separate file using the names of the two players and saves the result of each completed game. This allows previous game records to be displayed even after completing multiple games.

The project is designed to be simple and easy to understand while demonstrating how different Python functions can be combined to create a complete working application. Separate functions are used for important tasks such as creating and displaying the board, accepting player moves, checking winning conditions, checking draws, maintaining scores, saving game records, and displaying game history.

Main Features

- Two-player Tic Tac Toe game
- Player name input and validation
- X and O symbols for the players
- 3 × 3 game board
- Position selection from 1 to 9
- Invalid input handling
- Detection of horizontal, vertical, and diagonal wins
- Draw detection
- Score tracking
- Total games and draw count
- Game history

- File handling for storing game results
- Player details display
- Reset score and game records option
- Menu-driven interface
- Simple console-based user interface

Objective

The objective of this project is to build a functional Tic Tac Toe game using fundamental Python concepts and to understand how functions, loops, conditions, lists, user input, and file handling can work together in a complete Python application.

Technologies Used

- **Python**
- **Visual Studio Code**
- **Text File Handling**

Conclusion

This project provides practical experience in developing a small Python application from the beginning. It demonstrates the implementation of game logic, user input validation, score management, functions, and persistent game records using file handling.

The project can also be extended in the future with features such as a single-player mode, computer opponent, improved graphical interface, and additional game statistics.